


Course Code : 2301CS503

Date : 10-10-2025

Course Name : Data Mining

Duration : 150 Minutes

Total Marks : 70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicate maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 (A) Define Data Mining. Explain why it is crucial technology in today's data-driven world. **4**

(B) List and briefly describe three major issues in Data Mining **3**

OR

Explain different types of data attributes with examples.

(C) Explain the complete KDD (Knowledge Discovery in Databases) process with a neat diagram. How does Data Mining fit into this process? **7**

OR

Describe various data visualization techniques and discuss their importance in the initial phase of data analysis.

Q.2 (A) Why is data preprocessing an important step? List the major tasks in data preprocessing. **4**

(B) Explain data cleaning. List out methods/approaches to handle missing values. **3**

OR

What is data integration and what challenges are faced during this process?

(C) What is dimensionality reduction, and which method can be used for it? **7**

OR

Explain Data Transformation techniques in brief. Explain the following methods:

a) Normalization b) Discretization.

Q.3 (A) Define support and confidence in the context of association rule mining. Explain their significance. **4**

(B) What makes a pattern 'interesting'? Discuss objective measures for pattern evaluation. **3**

OR

Define association rule mining. What is market-basket analysis?

(C) For the given transaction data, find all frequent itemset using the Apriori algorithm. Use a minimum support count of 2. Then, generate strong association rules with a minimum confidence of 70%. **7**

TID	Items
1	{Bread, Milk}
2	{Bread, Diaper, Beer, Eggs}
3	{Milk, Diaper, Beer, Coke}
4	{Bread, Milk, Diaper, Beer}
5	{Bread, Milk, Diaper, Coke}

OR

Explain the FP-Growth algorithm for frequent itemset mining. How is it more efficient than the Apriori algorithm?

- Q.4 (A)** What is classification in data mining? Differentiate between classification and prediction. **4**
- (B)** Explain the working principle of the Naïve Bayes classifier. **3**

OR

Explain how a decision tree is constructed. Discuss the role of Information Gain in attribute selection for a decision tree.

- (C)** What is Cross Validation? Explain significance of it. Discuss different methods or strategies of it. **7**

OR

Describe the process of model evaluation in classification. Explain the concepts of a confusion matrix, accuracy, precision, and recall

- Q.5 (A)** What is Cluster Analysis? Differentiate it from classification **4**
- (B)** Briefly explain the K-Means partitioning method. **3**

OR

Differentiate between hierarchical and partitioning clustering methods.

- (C)** Explain the DBSCAN algorithm for density-based clustering. What are its advantages over K-Means? **7**

OR

What is an outlier? Describe different methods for outlier detection with suitable examples.
